Ontological Commitment, Representational Exclusion, and Reality Drift

How a Data Model Decides What Is Allowed to Exist

Knowledge Systems and Reality Drift - Note #2 v1

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The Basic Pattern

Every formal knowledge system begins by deciding what kinds of things exist. A data model may recognize patients, diagnoses, treatments, and outcomes. A commercial system may recognize customers, products, transactions, and accounts. A knowledge graph may recognize people, organizations, events, places, and relations.

These decisions are ontological commitments. They are not merely labels attached after observation. They determine what the system can store, compare, retrieve, infer, and act upon. A phenomenon that has no available class, property, or relationship may remain visible to a person while being absent from the operational world of the system.

Ontological commitment is unavoidable. The failure begins when a useful conceptualization is treated as a complete inventory of reality. The model stops functioning as one way of organizing the domain and begins functioning as the boundary of what the institution can know.

When a Model Becomes a World

Ontology engineering often evaluates a model through consistency, coverage, reuse, and competency questions: can the ontology answer the questions it was designed to answer? These are necessary tests, but the chosen questions already encode a view of what matters.

Once software, forms, reporting, access rules, and automated decisions depend on the ontology, its commitments become causal. People learn which facts the system accepts. Workers translate ambiguous cases into recognized fields. Unmodeled relationships disappear from dashboards and therefore from institutional attention.

The ontology has not become reality in a philosophical sense. It has become the actionable world available to the organization. Representational exclusion occurs when a condition is not necessarily denied, but cannot travel through the system in a form that produces recognition or response.

How Representational Exclusion Emerges

Stage 1 - Selection
Designers select the entities, relations, and distinctions needed for a defined purpose.

Stage 2 - Encoding
The conceptualization becomes a schema, form, database, taxonomy, or knowledge graph.

Stage 3 - Operationalization
Workflows and decisions are coupled to the encoded structure. What can be represented becomes easier to process.

Stage 4 - Adaptation
Users reshape descriptions and behavior to fit available categories. Exceptions are hidden in notes or lost during normalization.

Stage 5 - Exclusion
The system treats representational absence as practical irrelevance. Reality outside the ontology remains present but becomes institutionally weak.

Reality Drift and Ontological Authority

Reality Drift becomes possible when the ontology continues supporting efficient operation while its exclusions become consequential. The model appears successful because it answers the questions it was designed to answer. It may never reveal that the more important questions were excluded at design time.

Ontological authority is the power of a conceptual structure to determine which claims can become actionable. This authority grows as more systems reuse the same schema. Shared standards improve interoperability, but they also distribute the same omissions across institutions.

A reality-constrained ontology therefore needs more than formal validity. It needs mechanisms for contesting its commitments: residual categories that remain visible, provenance for modeling decisions, competency questions that test excluded cases, and revision channels accessible to the people represented.

The Representation Stack

Ontological commitments sit low in the representation stack. They shape later data before measurement or optimization begins. Experience becomes a record only through available fields; records become categories only through available classes; categories become metrics; metrics become decisions.

This explains why some failures cannot be repaired by improving an algorithm at the top of the stack. If the ontology never represented a relevant distinction, downstream models cannot recover it from formally clean but structurally incomplete data.

Examples Across Systems

Healthcare: A record system can represent diagnoses and procedures more easily than uncertainty, care context, or a patient's changing experience.

Identity Systems: Administrative models may require one stable name, address, household, or legal status even when lived identity is more complex.

Organizations: Customer relationship systems recognize interactions that fit predefined stages while informal trust, hesitation, and context remain invisible.

Artificial Intelligence: A graph can reason accurately over encoded relations while failing on facts that require a relationship its ontology never permitted.

Conclusion

Ontological commitment is the hidden beginning of many representational failures. Before a system measures badly or retrieves incorrectly, it has already decided what can count as an entity, a property, a relation, and a relevant question.

The central safeguard is not an ontology without commitments. Such a thing is impossible. It is an ontology that remains answerable to what it excludes. What cannot enter the model must retain a path through which it can challenge the model.

Related Phrases and Concepts

- ontological commitment
- conceptualization
- domain ontology
- upper ontology
- competency questions
- representational exclusion
- schema bias
- knowledge representation

Semantic Fidelity Project Resources

- Research Library (GitHub)
- Articles & Essays (Substack)
- Semantic Fidelity Glossary